3.4.1 i) Confirm formula $\operatorname{div}(\wp_i'/\wp_i)$

$$= -(0+\Lambda_i) + (1/2 + \Lambda_i) + (i/2 + \Lambda_i) - ((1+i)/2 + \Lambda_i).$$

by Ex 1.4.3.

Pf: $\wp_i$ has double pole at 0 — two simple zeros at the half-periods: $\frac{1}{2}, \frac{i}{2}, \frac{1+i}{2}$

$\wp_i'$ has a triple pole at 0 — simple zeros at

Thus $\operatorname{div}(\wp_i) = 2\left(\frac{1+i}{2} + \Lambda_i\right) - 2(0+\Lambda_i)$

$$\operatorname{div}(\wp_i') = \left(\frac{1+i}{2} + \Lambda_i\right) + (1/2 + \Lambda_i) + (i/2 + \Lambda_i) - 3(0 + \Lambda_i).$$

Thus $\operatorname{div}(\wp_i'/\wp_i) = \operatorname{div}(\wp_i') - \operatorname{div}(\wp_i)$ as above.

ii) What is $\operatorname{div}(\wp_{\mu_3}'/\wp_{\mu_3})$?

Sol: $\wp_{\mu_3}$ has double pole at 0 — one double zero at half-period $w = \frac{1+\mu_3}{3}$

$\wp_{\mu_3}'$ has triple pole at 0 — simple zeros at 3 half periods $w, \mu_3 w, \mu_3^2 w$

Thus $\operatorname{div}(\wp_{\mu_3}) = 2(w + \Lambda_{\mu_3}) - 2(0 + \Lambda_{\mu_3})$

$$\operatorname{div}(\wp_{\mu_3}') = (w + \Lambda_{\mu_3}) + (\mu_3 w + \Lambda_{\mu_3}) + (\mu_3^2 w + \Lambda_{\mu_3}) - 3(0 + \Lambda_{\mu_3}).$$

Thus $\operatorname{div}(\wp_i'/\wp_i) = \operatorname{div}(\wp_i') - \operatorname{div}(\wp_i)$

$$= -(0+\Lambda_{\mu_3}) - (w+\Lambda_{\mu_3}) + (\mu_3 w + \Lambda_{\mu_3}) + (\mu_3^2 w + \Lambda_{\mu_3}).$$

3.4.2 Show $\nu_x(f_1+f_2) \geq \min\{\nu_x(f_1), \nu_x(f_2)\}$. $f_1, f_2 \in \mathbb{C}(X)$, where x any point of compact Riemann surface X.

Pf: Choose local holomorphic coordinate z centered at x, so $z(x)=0$.

By definition, $f_i(z) = z^{\nu_x(f_i)} u_i(z)$ $i=1,2$.

where u_i meromorphic near 0 and $u_i(0) \neq 0$.

WLOG $\nu_x(f_1) \leq \nu_x(f_2)$, so

$$f_1(z)+f_2(z) = z^{\nu_x(f_1)}\left(u_1(z) + z^{\nu_x(f_2)-\nu_x(f_1)} u_2(z)\right)$$

$$\Rightarrow \nu_x(f_1+f_2) \geq \nu_x(f_1) = \min\{\nu_x(f_1), \nu_x(f_2)\}.$$

3.4.3. Recall the discussion after Thm 3.3.1,

$$\Omega^1(X(\mathbb{P})) = \mathbb{C}(X(\mathbb{P}))\lambda \quad \forall \lambda \in \Omega^1(X(\mathbb{P})).$$

Show $\Omega^1(X(\mathbb{P})) \to \mathbb{C}(X(\mathbb{P}))$ $f_0\lambda \mapsto f_0$ is isomorphism of vectorspaces $\Omega^1_{hol}(X(\mathbb{P})) \xrightarrow{\sim} L(\lambda)$.

Pf: well-defined: $\Omega^{\otimes k/2}(X(\mathbb{P})) = \mathbb{C}(X(\mathbb{P}))\,w(t)$

implies $\Omega^1(X(\mathbb{P})) = \mathbb{C}(X(\mathbb{P}))\lambda$.

Thus $\Phi : \Omega^1(X(\mathbb{P})) \to \mathbb{C}(X(\mathbb{P}))$ is well-defined, linear, and bijective with inverse $f\lambda \leftarrow\!\shortmid f$.

Holomorphy: Let $w = f_0\lambda \in \Omega^1(X(\mathbb{P}))$. $\forall x \in X(\mathbb{P})$

$$\nu_x(w) = \nu_x(f_0) + \nu_x(\lambda).$$

Hence w holomorphic at $x \Leftrightarrow \nu_x(f_0) \geq -\nu_x(\lambda)$.

exactly $f_0 \in L(\lambda)$ the Riemann-Roch space.

Restriction: therefore $w \in \Omega^1_{hol}(X(\mathbb{P})) \Leftrightarrow f_0 \in L(\lambda)$.

$$\Phi\big|_{\Omega^1_{hol}(X(\mathbb{P}))} : \Omega^1_{hol}(X(\mathbb{P})) \xrightarrow{\sim} L(\lambda)$$
$$w \mapsto f_0$$